

## Part1

Independent Variable: Race

dependent Variable: Highest year of school completed

H0: The mean highest year of school completed is equal across racial groups

H1: The mean highest year of school completed is not equal across racial groups

### Descriptives

Highest year of school completed

|       | N    | Mean  | Std. Deviation | Std. Error | 95% Confidence Interval for Mean |             | Minimum | Maximum |
|-------|------|-------|----------------|------------|----------------------------------|-------------|---------|---------|
|       |      |       |                |            | Lower Bound                      | Upper Bound |         |         |
| WHITE | 1692 | 13.98 | 2.787          | .068       | 13.85                            | 14.11       | 0       | 20      |
| BLACK | 384  | 13.37 | 2.724          | .139       | 13.10                            | 13.65       | 0       | 20      |
| OTHER | 269  | 12.68 | 4.012          | .245       | 12.20                            | 13.17       | 0       | 20      |
| Total | 2345 | 13.73 | 2.974          | .061       | 13.61                            | 13.85       | 0       | 20      |

### Tests of Homogeneity of Variances

|                                  |                                      | Levene Statistic | df1 | df2      | Sig.  |
|----------------------------------|--------------------------------------|------------------|-----|----------|-------|
| Highest year of school completed | Based on Mean                        | 21.216           | 2   | 2342     | <.001 |
|                                  | Based on Median                      | 21.087           | 2   | 2342     | <.001 |
|                                  | Based on Median and with adjusted df | 21.087           | 2   | 1931.436 | <.001 |
|                                  | Based on trimmed mean                | 21.198           | 2   | 2342     | <.001 |

### ANOVA

Highest year of school completed

|                | Sum of Squares | df   | Mean Square | F      | Sig.  |
|----------------|----------------|------|-------------|--------|-------|
| Between Groups | 449.078        | 2    | 224.539     | 25.921 | <.001 |
| Within Groups  | 20287.205      | 2342 | 8.662       |        |       |
| Total          | 20736.283      | 2344 |             |        |       |

### ANOVA Effect Sizes<sup>a</sup>

|                                  |                             |      | 95% Confidence Interval |       |
|----------------------------------|-----------------------------|------|-------------------------|-------|
| Point Estimate                   |                             |      | Lower                   | Upper |
| Highest year of school completed | Eta-squared                 | .022 | .011                    | .034  |
|                                  | Epsilon-squared             | .021 | .010                    | .033  |
|                                  | Omega-squared Fixed-effect  | .021 | .010                    | .033  |
|                                  | Omega-squared Random-effect | .011 | .005                    | .017  |

a. Eta-squared and Epsilon-squared are estimated based on the fixed-effect model.

## Robust Tests of Equality of Means

Highest year of school completed

|       | Statistic <sup>a</sup> | df1 | df2     | Sig.  |
|-------|------------------------|-----|---------|-------|
| Welch | 18.635                 | 2   | 523.407 | <.001 |

a. Asymptotically F distributed.

Result:.

Levene's Test of Homogeneity of Variances was significant, ( $p < 0.001$ ), Because the homogeneity of variance assumption was not met, Although the standard ANOVA was significant ( $F=25.921$ ,  $p < 0.001$ ). So Welch's ANOVA was used. Welch's  $F= 18.635$ ,  $p < 0.001$ . Therefore, the null hypothesis was rejected.

Descriptive statistics showed that White respondents had the highest mean years of schooling ( $M = 13.98$ ,  $SD = 2.79$ ), followed by Black respondents ( $M = 13.37$ ,  $SD = 2.72$ ), and Other respondents ( $M = 12.68$ ,  $SD = 4.01$ )

In conclusion, race was associated with statistically significant differences in educational attainment, with White respondents reporting the highest average years of schooling, followed by Black respondents, and then Other respondents.

Test Assumptions:

A one-way ANOVA was appropriate for this analysis because only three independent groups were compared (White vs. Black vs. Other) on a continuous dependent variable (Highest year of school completed). The dependent variable was assumed to be approximately normally distributed within each group. Finally, Levene's Test was significant ( $p < 0.001$ ), therefore, welch's test was used.

## Multiple Comparisons

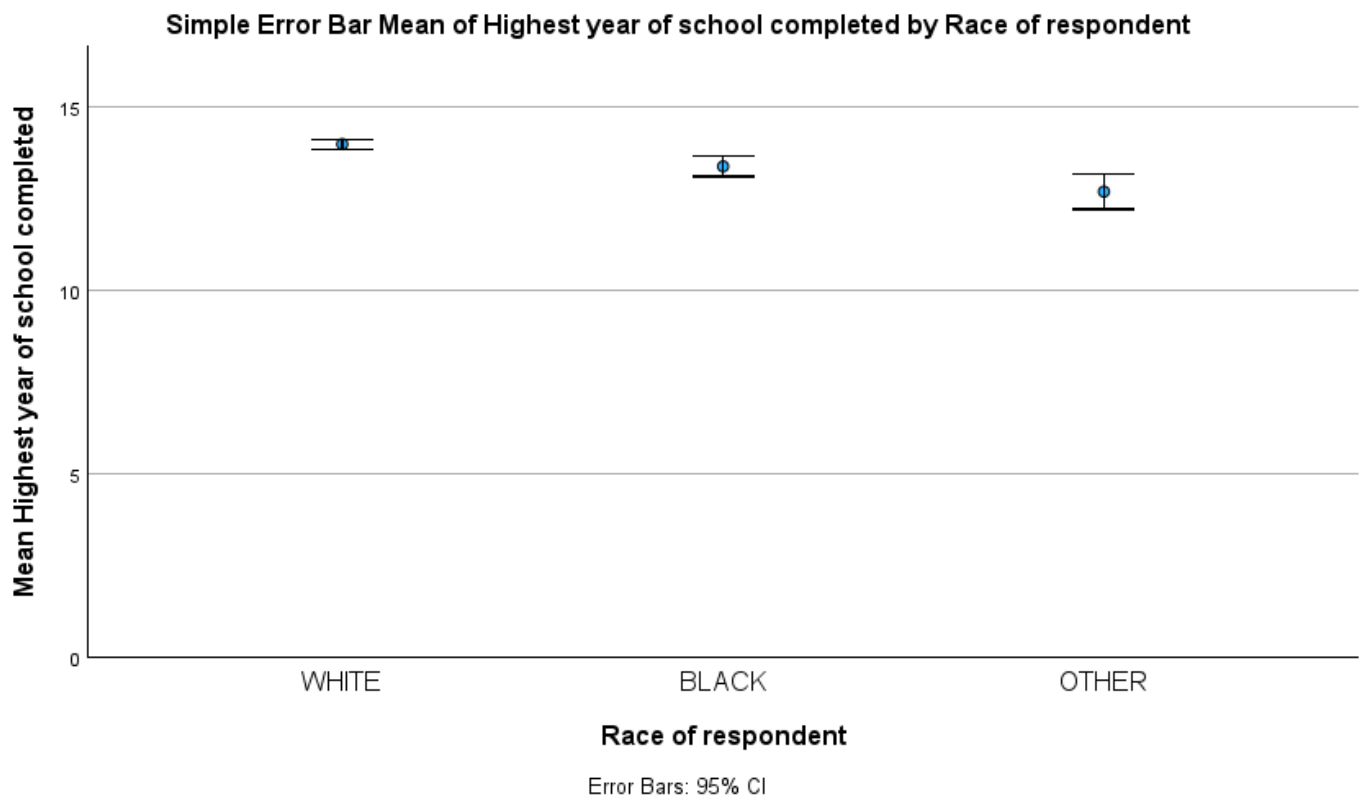
Dependent Variable: Highest year of school completed

Bonferroni

| (I) Race of respondent | (J) Race of respondent | Mean Difference (I-J) | Std. Error | Sig.  | 95% Confidence Interval |             |
|------------------------|------------------------|-----------------------|------------|-------|-------------------------|-------------|
|                        |                        |                       |            |       | Lower Bound             | Upper Bound |
| WHITE                  | BLACK                  | .608 <sup>*</sup>     | .166       | <.001 | .21                     | 1.01        |
|                        | OTHER                  | 1.296 <sup>*</sup>    | .193       | <.001 | .83                     | 1.76        |
| BLACK                  | WHITE                  | -.608 <sup>*</sup>    | .166       | <.001 | -1.01                   | -.21        |
|                        | OTHER                  | .688 <sup>*</sup>     | .234       | .010  | .13                     | 1.25        |
| OTHER                  | WHITE                  | -1.296 <sup>*</sup>   | .193       | <.001 | -1.76                   | -.83        |
|                        | BLACK                  | -.688 <sup>*</sup>    | .234       | .010  | -1.25                   | -.13        |

\*. The mean difference is significant at the 0.05 level.

Bonferroni post hoc comparisons indicated that all pairwise comparisons were statistically significant. White respondents had significantly higher educational attainment than Black respondents (mean difference = 0.608,  $p < 0.001$ ) and Other respondents (mean difference = 1.296,  $p < 0.001$ ). Black respondents also had significantly higher educational attainment than Other respondents (mean difference = 0.688,  $p = 0.010$ ).



The error bar chart visually supports the statistical findings. White respondents had the highest mean years of schooling, followed by Black respondents, while respondents in the Other category had the lowest mean educational attainment. The 95% confidence intervals show clear separation among the group means, suggesting meaningful differences across racial groups. This visual pattern is consistent with the Welch ANOVA and Bonferroni post hoc results, which indicated statistically significant differences between all three racial groups.

The White group has the narrowest confidence interval, likely because it has a much larger sample size ( $n = 1,692$ ) than the Black group ( $n = 384$ ) and the Other group ( $n = 269$ ).

## Part 2

The purpose of this One-Way ANOVA exercise was to determine whether the mean highest year of school completed differed significantly across racial groups (White, Black, and Other). One-Way ANOVA compares group means by calculating an F-ratio, which is the ratio of variance between groups to variance within groups. In this analysis, Levene's Test of Homogeneity of Variances was significant ( $p < .001$ ), indicating unequal variances, so Welch's ANOVA was used as an alternative. The Welch test showed a statistically significant difference in educational attainment among racial groups, Welch's  $F = 18.64$ ,  $p < .001$ , leading to rejection of the null hypothesis. Post hoc comparisons further indicated that all three racial groups differed significantly from one another, with White respondents reporting the highest average years of schooling, followed by Black respondents, and then Other respondents. Overall, the results suggest that race is associated with statistically significant differences in educational attainment in this sample.